

Exercises II.2

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2.1. One can associate an $n \times n$ matrix M_σ with a permutation $\sigma \in S_n$, by letting the entry at $(i, (i)\sigma)$ be 1 and letting all other entries be 0. For example, the matrix corresponding to the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \in S_3$$

would be

$$M_\sigma = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Prove that, with this notation,

$$M_{\sigma r} = M_\sigma M_r$$

for all $\sigma, r \in S_n$, where the product on the right is the ordinary product of matrices.

Proof. First, observe that each M_σ has exactly one nonzero element in each row and each column. For rows, this is because σ is a function, so $(i)\sigma$ has exactly one value. For columns, there must be at least one element in each column because σ is surjective, and since σ is injective, there cannot be two or more entries since that would imply $(i_1)\sigma = (i_2)\sigma$ for some $i_1 \neq i_2$.

We clarify that σr applies σ *before* r and $M_\sigma M_r$ applies M_σ *after* M_r . We denote the entry in row j , column k of M_σ by $a_{j,k}$ and the same in M_r by $b_{j,k}$. By definition, the entry in row j , column k in $M_\sigma M_r$ is

$$\sum_{m=1}^n a_{j,m} b_{m,k},$$

which has nonzero terms when $(j)\sigma = m$ and $(m)r = k$. Thus the only nonzero (and thus, valued 1) entries are those where $((j)\sigma)r = k$, which is exactly the definition of the nonzero entries of $M_{\sigma r}$. $\square$

2.2. Prove that if $d \leq n$, then S_n contains elements of order d .

Proof. We proceed by induction over n . Clearly the conclusion holds for S_1 since the identity has order 1. Thus, assume that, for some n , the theorem holds for S_{n-1} . Let $\sigma_d \in S_{n-1}$ be an element of order d for each $d \leq n-1$. Then the permutation $f_d \in S_n$ defined by

$$f_d(x) = \begin{cases} \sigma_d(x) & x < n \\ x & x = n \end{cases}$$

has order d since if $x < n$, then $f_d^d(x) = \sigma_d^d(x) = x$ (and for some such x , this holds for no value less than d) by definition of σ_d , and if $x = n$, then obviously $f_d^d(x) = x$.

If $d = n$, then we show that the “rightward shift” permutation is of order d . Indeed, define f_n by

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ 2 & 3 & \cdots & 1 \end{pmatrix},$$

and we see that $f_n^n(x) = x$ for each x , but this holds for no other value less than n , so $|f_n| = n$. $\square$

2.3. For every positive integer n find an element of order n in $S_{\mathbb{N}}$.

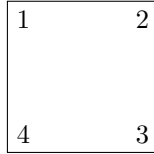
Proof. Define $\sigma_n \in S_{\mathbb{N}}$ by

$$\sigma_n(x) = \begin{cases} (x+1) \bmod n & 0 \leq x < n \\ x & x < 0 \text{ or } x \geq n \end{cases}$$

That this has order n is a feature of elementary modular arithmetic. $\square$

2.4. Define a homomorphism $D_8 \rightarrow S_4$ by labelling vertices of a square, as we did for a triangle in §2.2. List the 8 permutations in the image of this homomorphism.

Proof. We label the square like so:



so that the symmetries produce obvious permutations of the set $\{1, 2, 3, 4\}$. Namely, the “pure” rotations are the following:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}.$$

These correspond to (or “are mapped to by”) the clockwise rotations 0° , 90° , 180° , and 270° respectively. If we let r be the 90° clockwise rotation, then these correspond to r^0, r^1, r^2 , and r^3 respectively. Also, we can flip along the following

lines: northwest to southeast, southwest to northeast, west to east, and finally north to south. These correspond to the following permutations (respectively):

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}.$$

□

2.5. Describe generators and relations for all dihedral groups D_{2n} . (Hint: Let x be the reflection about a line through the center of a regular n -gon and a vertex, and let y be the counterclockwise rotation by $2\pi/n$. The group D_{2n} will be generated by x and y , subject to three relations. To see that these relations really determine D_{2n} , use them to show that any product $x^{i_1}y^{i_2}x^{i_3}y^{i_4}\dots$ equals x^iy^j for some i, j with $0 \leq i \leq 1$, $0 \leq j < n$.)

Proof. Let e be the identity, x the reflection about a line through the center of a regular n -gon and one of its vertices, and y the clockwise rotation by $2\pi/n$. These are paired with the obvious relations $x^2 = e$ and $y^n = e$, along with the relation $yx = xy^{n-1}$.

To prove this, we use induction to show that any product $x^{i_1}y^{i_2}x^{i_3}y^{i_4}\dots$ for integers i_1, i_2, i_3, i_4 equals x^iy^j for some i, j with $0 \leq i \leq 1$ and $0 \leq j < n$. It will suffice to show this for all nonnegative integers, since any product including negative exponents can be converted into one including only nonnegative exponents using the relations $x^{-1} = x$ and $y^{-1} = y^{n-1}$, which derive from the proposed generating relations. Given $i_1, i_2 \in \mathbb{Z}^{\geq 0}$ (our base case),

$$x^{i_1}y^{i_2} = x^{i_1 \bmod 2}y^{i_2 \bmod n}$$

since $x^2 = e$ and $y^n = e$. Of course, $0 \leq i_1 \bmod 2 \leq 1$ and $0 \leq i_2 \bmod n < n$, so our base case holds.

We now prove a small lemma: for any $i \in \mathbb{Z}^{\geq 0}$ and $j \in \{0, 1\}$,

$$y^ix^j = x^jy^i \tag{1}$$

for some $k \in \mathbb{Z}^{\geq 0}$. First, observe that if $i = 0$ or $j = 0$, (1) is obvious, so we can assume $i > 0$ and $j = 1$. We proceed by induction over i . The case $i = 1$ is already given since $yx = xy^{n-1}$. If $i > 1$ and (1) holds for $i - 1$, Then

$$\begin{aligned} y^ix &= y^{i-1}yx \\ &= y^{i-1}xy^{n-1} \\ &= xy^ky^{n-1} \\ &= xy^{k+n-1} \end{aligned}$$

for some $k \in \mathbb{Z}^{\geq 0}$. Thus (1) holds for all nonnegative integers i .

Now, assume that, for some m , $x^{i_1}y^{i_2}\dots x^{i_{2m-3}}y^{i_{2m-2}} = x^iy^j$ for some i, j with $0 \leq i \leq 1$ and $0 \leq j < n$. Then for some $j_1, j_2, k_1, k_2, k_3 \in \mathbb{Z}^{\geq 0}$ such that

each $0 \leq j_\alpha \leq 1$ and each $0 \leq k_\beta < n$,

$$\begin{aligned} x^{i_1} y^{i_2} \dots x^{i_{2m-1}} y^{i_{2m}} &= x^{j_1} y^{k_1} x^{i_{2m-1}} y^{i_{2m}} \\ &= x^{j_1} y^{k_1} x^{j_2} y^{k_2} \\ &= x^{j_1} x^{j_2} y^{k_3} y^{k_2} \\ &= x^{j_1+j_2} y^{k_3+k_2}, \end{aligned}$$

leaving us with our already-proven base case. $\square$

2.6. For every positive integer n construct a group containing elements g, h such that $|g| = 2$, $|h| = 2$, and $|gh| = n$. (Hint: For $n > 1$, D_{2n} will do.)

Proof. For $n = 1$, any group with an element g of order 2 (e.g. S_2) satisfies $|g^2| = |e| = 1$. For $n > 1$, consider D_{2n} with generators and relations as described in the previous exercise. Then

$$\begin{aligned} (yx)^2 &= yxyx \\ &= yxyy^{n-1} \\ &= yy^{n-1} \\ &= y^n \\ &= e, \end{aligned}$$

so $|yx| = 2$. Thus letting $g = yx$ and $h = x$ satisfies the desired property since $|gh| = |yxx| = |y| = n$. $\square$

2.7. Find all elements of D_{2n} that commute with every other element. (The parity of n plays a role.)

Proof. If n is odd, then the only element that commutes with every other element seems to be e . If n is even, however, then we have the case of

$$\begin{aligned} y^{n/2}x &= xy^{(n-1)(n/2)} \\ &= xy^{n^2/2-n/2} \\ &= xy^{(n^2/2-n/2) \bmod n} \\ &= xy^{(n^2/2) \bmod n - (n/2) \bmod n} \\ &= xy^{-(n/2) \bmod n} \\ &= xy^{n/2}, \end{aligned}$$

so $y^{n/2}$ commutes with x , and given any element of D_{2n} expressed as $x^i y^j$ where $0 \leq i \leq 1$ and $0 \leq j < n$ as in Exercise 2.5, we have

$$\begin{aligned} y^{n/2}x^i y^j &= x^i y^{n/2} y^j \\ &= x^i y^{n/2+j} \\ &= x^i y^{j+n/2} \\ &= x^i y^j y^{n/2}, \end{aligned}$$

so $y^{n/2}$ commutes with every element, as does e . □

2.8. Find the orders of the groups of symmetries of the five ‘platonic solids’.

Proof. UNFINISHED □

2.9. Verify carefully that ‘congruence mod n ’ is an equivalence relation.

Proof. Let $\sim$ be the relation “congruence mod n ”. Given $x, y, z \in \mathbb{Z}$, clearly $n|x - x$, so $\sim$ is reflexive. If $x \sim y$, then for some integer a , $y - x = an$, so $x - y = -an$, implying that $y \sim x$. Finally, if $x \sim y$ and $y \sim z$, let $y - x = an$ and $z - y = bn$ for some integers a, b . Then $z - x = (z - y) + (y - x) = bn + an = (b + a)n$, so $x \sim z$. □

2.10. Prove that $\mathbb{Z}/n\mathbb{Z}$ consists of precisely n elements.

Proof. Clearly $\mathbb{Z}/n\mathbb{Z}$ contains *at least* n elements, since given integers $j, k \in \{0, \dots, n-1\}$, $n|k - j$ if and only if $k = j$. If $x \in \mathbb{Z}$ and $x \geq n$ or $x < 0$, then there are unique $a, b \in \mathbb{Z}$ such that $0 \leq b < n$ and $x = an + b$, and $a \neq 0$, so $x - b = an$, meaning $n|x - b$ and $b \equiv x \pmod{n}$. Thus $[b]_n = [x]_n$, so there are no more than n elements of $\mathbb{Z}/n\mathbb{Z}$, and thus $|\mathbb{Z}/n\mathbb{Z}| = n$. □

2.11. Prove that the square of every odd integer is congruent to 1 modulo 8.

Proof. We proceed by induction. Obviously $1^2 = 1$ is congruent to 1 modulo 8, so let n be some odd integer such that $(n-2)^2 \equiv 1 \pmod{8}$. We can write $8a = (n-2)^2 - 1$ and $n-1 = 2b$, where a and b are integers. Then

$$\begin{aligned} n^2 - 1 &= ((n-2) + 2)^2 - 1 \\ &= (n-2)^2 + 4(n-2) + 4 - 1 \\ &= 8a + 1 + 4(n-1) - 1 \\ &= 8a + 4(2b) \\ &= 8(a+b), \end{aligned}$$

so $n^2 \equiv 1 \pmod{8}$. □

2.12. Prove that there are no nonzero integers a, b, c such that $a^2 + b^2 = 3c^2$. (Hint: By studying the equation $[a]_4^2 + [b]_4^2 = 3[c]_4^2$ in $\mathbb{Z}/4\mathbb{Z}$, show that a, b, c would all have to be even. Letting $a = 2k, b = 2\ell, c = 2m$, you would have $k^2 + \ell^2 = 3m^2$. What’s wrong with that?)

Proof. Note that

$$\begin{aligned} [0]_4^2 &= [0]_4 \\ [1]_4^2 &= [1]_4 \\ [2]_4^2 &= [0]_4 \\ [3]_4^2 &= [1]_4. \end{aligned}$$

Let a, b, c be nonzero integers such that $a^2 + b^2 = 3c^2$. Then $[a]_4^2$ is either $[0]_4$ or $[1]_4$ since those are the only possible values of squared elements in $\mathbb{Z}/4\mathbb{Z}$, and likewise for $[b]_4^2$ and $[c]_4^2$. Thus $3[c]_4^2$ is either $[0]_4$ or $[3]_4$. It cannot be $[3]_4$, however, since any assignment of $[0]_4$ or $[1]_4$ for $[a]_4^2$ and $[b]_4^2$ sums to at most $[2]_4$. Thus $3[c]_4^2 = [0]_4$, so $[c]_4^2 = [0]_4$, which forces $[a]_4^2 = [b]_4^2 = [0]_4$ since they must add to $[0]_4$. Given this fact, together with the note at the beginning of this proof, $[a]_4$, $[b]_4$, and $[c]_4$ must be either $[0]_4$ or $[2]_4$. Therefore, a , b , and c are all even.

Let $a = 2k$, $b = 2\ell$, and $c = 2m$. Then

$$\begin{aligned} a^2 + b^2 &= c^2 \\ \implies 4k^2 + 4\ell^2 &= 4m^2 \\ \implies k^2 + \ell^2 &= m^2. \end{aligned}$$

If k , ℓ , and m are all even, there is no issue yet, but we can apply this procedure of dividing a , b , and c by 2 until an odd number is reached, which would then contradict our finding that all three bases must be even. $\square$

2.13. Prove that if $\gcd(m, n) = 1$, then there exist integers a and b such that

$$am + bn = 1.$$

(Use Corollary 2.5.) Conversely, prove that if $am + bn = 1$ for some integers a and b , then $\gcd(m, n) = 1$.

Proof. By Corollary 2.5, $[m]_n$ generates $\mathbb{Z}/n\mathbb{Z}$, so there exists an integer a such that $a[m]_n = [am]_n = [1]_n$. Thus, by definition of $\mathbb{Z}/n\mathbb{Z}$, $n \mid 1 - am$, so there exists an integer b such that $bn = 1 - am$.

Conversely, if $am + bn = 1$ for integers a and b , then $bn = 1 - am$, so $n \mid 1 - am$, meaning $[1]_n = [am]_n = a[m]_n$ in $\mathbb{Z}/n\mathbb{Z}$. Thus given any $[\ell]_n \in \mathbb{Z}/n\mathbb{Z}$, $[\ell]_n = \ell[1]_n = \ell a[m]_n$, so $[m]_n$ generates $\mathbb{Z}/n\mathbb{Z}$. By Corollary 2.5, then, $\gcd(m, n) = 1$. $\square$

2.14. State and prove an analog of Lemma 2.2, showing that the multiplication on $\mathbb{Z}/n\mathbb{Z}$ is a well-defined operation.

Proof. We need to show that if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then

$$ab \equiv a'b' \pmod{n}.$$

Let $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$. Then $xn = a' - a$ and $yn = b' - b$ for some integers x and y . Thus

$$\begin{aligned} a'b' - ab &= (xn + a)(yn + b) - ab \\ &= xyn^2 + ayn + bxn + ab - ab \\ &= n(xyn + ay + bx), \end{aligned}$$

so $ab \equiv a'b' \pmod{n}$ as desired. $\square$

2.15. Let $n > 0$ be an odd integer.

- Prove that if $\gcd(m, n) = 1$, then $\gcd(2m + n, 2n) = 1$. (Use Exercise 2.13.)
- Prove that if $\gcd(r, 2n) = 1$, then $\gcd(\frac{r+n}{2}, n) = 1$. (Ditto.)
- Conclude that the function $[m]_n \rightarrow [2m + n]_{2n}$ is a bijection between $(\mathbb{Z}/n\mathbb{Z})^*$ and $(\mathbb{Z}/2n\mathbb{Z})^*$.

The number $\phi(n)$ of elements of $(\mathbb{Z}/n\mathbb{Z})^*$ is *Euler's ϕ -function*. The reader has just proved that if n is odd, then $\phi(2n) = \phi(n)$. Much more general formulas will be given later on.

Proof.

- Since n is odd and $\gcd(m, n) = 1$, $\gcd(2m, n) = 1$ as well because no factor of n divides m and 2 is not a factor of n . Therefore, $[2m]_n = [2m + n]_n$ generates $\mathbb{Z}/n\mathbb{Z}$, so $\gcd(2m + n, n) = 1$. Finally, since $2m$ is even and n is odd, $2m + n$ is odd, so by the same logic we started with, $\gcd(2m + n, 2n) = 1$.
- Since $\gcd(r, 2n) = 1$, r must be odd, so $\gcd(r, n) = 1$. This means that $[r]_n = [r + n]_n$ generates $\mathbb{Z}/n\mathbb{Z}$, so in fact $\gcd(r + n, n) = 1$. Finally, since both r and n are odd, $r + n$ must be even, so $\frac{r+n}{2}$ is an integer and we get $\gcd(\frac{r+n}{2}, n) = 1$.
- Let $[m]_n \in (\mathbb{Z}/n\mathbb{Z})^*$. Then, by definition, $\gcd(m, n) = 1$, and we can define $f : (\mathbb{Z}/n\mathbb{Z})^* \rightarrow (\mathbb{Z}/2n\mathbb{Z})^*$ by $f([m]_n) = [2m + n]_{2n}$. First we show that this is well defined by letting $[m_1]_n = [m_2]_n$, so that $m_2 - m_1 = an$ for some integer a , and thus

$$(2m_2 + n) - (2m_1 + n) = 2(m_2 - m_1) = 2an,$$

which is divisible by $2n$ as desired.

Now assume that $f([m_1]_n) = f([m_2]_n)$ for some m_1, m_2 relatively prime to n . Then by definition there exists some integer a such that

$$an = (2m_2 + n) - (2m_1 + n) = 2(m_2 - m_1).$$

Thus n divides $m_2 - m_1$ (it cannot divide 2), so $[m_1]_n = [m_2]_n$ and f is injective.

Given $[m]_{2n} \in (\mathbb{Z}/2n\mathbb{Z})^*$, we know that $\gcd(m, 2n) = 1$, so by the second bullet point, $\gcd(\frac{m+n}{2}, n) = 1$. This helps because, in fact,

$$\begin{aligned} f\left(\left[\frac{m+n}{2}\right]_n\right) &= \left[2\left(\frac{m+n}{2}\right) + n\right]_{2n} \\ &= [m + 2n]_{2n} \\ &= [m]_{2n}. \end{aligned}$$

Thus f is bijective.

□

2.16. Find the last digit of $1238237^{18238456}$. (Work in $\mathbb{Z}/10\mathbb{Z}$.)

Proof. Exercise 2.14 implies that exponentiation is well-defined on $\mathbb{Z}/n\mathbb{Z}$ since

$$[a^b]_n = [\underbrace{a \cdots a}_{b \text{ times}}]_n = [\underbrace{[a]_n \cdots [a]_n}_{b \text{ times}}]_n = [a]_n^b.$$

Furthermore,

$$[a]_n^b = [a]_n^{(b \bmod |[a]_n|)}.$$

In the present case, of course $[1238237]_{10} = [7]_{10}$, and the order of $[7]_{10}$ is 4, as is easily checked. This allows us to conclude that

$$\begin{aligned} [1238237^{18238456}]_{10} &= [1238237]_{10}^{18238456} \\ &= [7]_{10}^{18238456} \\ &= [7]_{10}^{18238456 \bmod 4} \\ &= [7]_{10}^0 \\ &= [1]_{10}. \end{aligned}$$

Thus the last digit of $1238237^{18238456}$ is 1. □

2.17. Show that if $m \equiv m' \pmod{n}$, then $\gcd(m, n) = 1$ if and only if $\gcd(m', n) = 1$.

Proof. ($\implies$) Given $\gcd(m, n) = 1$, since $m \equiv m' \pmod{n}$ we have $an = m' - m$ for some integer a . Then we simply have

$$\begin{aligned} [m]_n &= [m + an]_n \\ &= [m + m' - m]_n \\ &= [m']_n. \end{aligned}$$

($\impliedby$) The opposite direction follows by the same logic, interchanging the roles of m' and m . □

2.18. For $d \leq n$, define an injective function $\mathbb{Z}/d\mathbb{Z} \rightarrow S_n$ preserving the operation, that is, such that the sum of equivalence classes in $\mathbb{Z}/d\mathbb{Z}$ corresponds to the product of the corresponding permutations.

Proof. UNFINISHED □

2.19. Both $(\mathbb{Z}/5\mathbb{Z})^*$ and $(\mathbb{Z}/12\mathbb{Z})^*$ consist of 4 elements. Write their multiplication tables, and prove that no re-ordering of the elements will make them match.

Proof. It is easy to see that

$$(\mathbb{Z}/5\mathbb{Z})^* = \{[1]_5, [2]_5, [3]_5, [4]_5\}$$

and

$$(\mathbb{Z}/12\mathbb{Z})^* = \{[1]_5, [5]_5, [7]_5, [11]_5\}.$$

For the former, we have the multiplication table (omitting the brackets and subscripts when denoting $[*]_5$)

	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

and for the later we have

	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

From this we can see that $\mathbb{Z}/5\mathbb{Z}$ has exactly one element of order 2, and in $\mathbb{Z}/12\mathbb{Z}$ three elements have order 2, so they cannot be the same. $\square$